HAMZA TARIQ

Erlangen, Germany | +49 160 92843294 | hamza1.htzs@gmail.com | linkedin.com/in/hamza-t-795a20218

PROFILE

M.Sc. student in Autonomy Technologies (FAU Erlangen-Nürnberg) with a mechanical engineering background and hands-on skills in Python Automation and Robotics. 1.5+ years of experience in ERP-based production planning at Changan Autos. Analysed operational data, built Excel/VBA and Power BI reporting used by top management and 5 departments, and coordinated a 19-person cross-functional team. Strong in turning complex data into clear, decision-ready materials, with a structured, proactive working style.

SKILLS

Programming, Data Analysis & Reporting: Python (pandas, numpy, matplotlib), C++, MATLAB, SQL (intermediate), Power BI (Power Query, KPI dashboards), advanced Excel (pivot tables, VBA macros, dynamic charts).
Microsoft 365 & Automation: Excel, PowerPoint, VBA-based workflow automation, Microsoft Copilot, Claude and ChatGPT for reporting and documentation.
Presentation & Communication: PowerPoint, structured KPI reporting for top management, simplifying technical topics for non-technical stakeholders
Coordination & Process Work: Cross-functional stakeholder coordination, consolidation of inputs from multiple teams, structured KPI reporting, process optimisation, root-cause and bottleneck analysis, JIT inventory planning, Microsoft Project ROI and cost modelling
ERP & Systems: SAP (basic), Oracle SCM/ASCP
Engineering Tools: SolidWorks, PTC Creo, ANSYS, COMSOL, CAD, Arduino, PLC

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) — Erlangen, Germany
M.Sc. Autonomy Technologies — *Apr 2026 – Apr 2028(Expected)*

- Focus: robotics, control engineering, Machine Learning, intelligent autonomous systems, computer vision; coursework includes machine learning, automation, deep leaning and signal processing.

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) — Topi, Pakistan
B.Sc. Mechanical Engineering — *Aug 2019 – Jun 2023*

- Final-year project: bat-inspired micro UAV (see Projects).

PROFESSIONAL EXPERIENCE

Changan Automobile – Master Changan Motors Ltd — Karachi, Pakistan
Assistant Manager – Production Planning & Control (SCM) — *Jul 2024 – Jan 2026*

- Coordinated short-term production planning for 4-8 assembly lines across 3 manufacturing areas with a 19-person cross-functional team using SAP and Oracle SCM; applied JIT principles to improve forecast accuracy and material availability by 30 %.
- Built Excel (VBA, pivot tables) and Power BI KPI dashboards for top-management review, cutting manual reporting errors by about 75 %; automated recurring reports with VBA and AI tools, saving about 10 hours per week across 5 departments.
- Analysed daily SAP and Oracle SCM data to detect deviations and bottlenecks, extending early warning for line stoppages from minutes to up to 2 days and informing procurement, logistics and production decisions.
- Resolved a DMS issue by restructuring legacy data and leading a working session with the Chinese Technical Director's team, saving an estimated 20+ working hours versus a rebuild.
- Avoided a line stoppage on a 260-unit government order during a tyre delivery delay by releasing vehicles without the missing part and fitting it on arrival.

Amreli Steels Limited — Karachi, Pakistan
Trainee Engineer – Utilities & Maintenance — *Mar 2024 – May 2024*

- Performed vibration analysis, thermography and oil analysis on rolling-mill equipment (EOT cranes, compressors, cooling towers) and scheduled the maintenance in a structured cycle, reducing downtime by 25 % and extending the life of moving parts by 10 %.

Heavy Industries Taxila (HIT) — Taxila, Pakistan
Design Engineering Intern – ARDIC — *Jul 2022 – Aug 2022*

- Contributed to the design and analysis of military tracked vehicles; created and modified mechanical components using CAD tools.

Engro Polymers and Chemicals Ltd. (EPCL) Karachi, Pakistan

Plant Operations Intern (Process Plant) *Jun 2021 – Aug 2021*

- Analysed plant energy consumption data and prepared a fact-based feasibility study for a solar plant, including supplier cost estimates and ROI modelling; projected a payback period of about 2 years.

PROJECTS

AUTODIDACT: Self-Improving Robot Learning Loop

- Built within 48 hours, in a team of five at the Siemens Healthineers Robotics & AI Hackathon (Erlangen, 2026, semi-finalist), an autonomous learning loop for a simulated cath-lab robot (MuJoCo twin of the FR3 Duo) that detects gaps in its training data, requests targeted demonstrations, retrains, and promotes a new policy only if a statistical test proves it is better.
- Implemented data validation and failure-analysis tooling, a safety filter with stop-and-resume that cut keep-out-zone violations from 24 to 0 of 40 runs, and a logged decision ledger, making every model update traceable.
- Ran the full cycle end to end without human intervention in 66 minutes, with a live dashboard showing data, runs and decisions.

Bat-Inspired Micro UAV – Final Year Project

- Designed a bio-inspired micro UAV in SolidWorks for disaster-response and surveillance, with a 4-DOF wing mechanism (flapping, morphing, twisting) replicating bat-wing aerodynamics, and 3D-printed lightweight components.
- Ran CFD/FEA in ANSYS, COMSOL and MATLAB and simulated wing dynamics in MSC Adams to optimize aerodynamics, structural integrity and energy efficiency.
- Built a microcontroller-based (Arduino/Raspberry Pi) control system for real-time wing adjustment, and validated a scaled prototype in wind tunnel and field tests.

Autonomous Firefighting Robot

- Engineered an autonomous fire-fighting robot system, covering sensing, control and actuation.

Wind Turbine Modelling:

- Modelled wind turbine systems using SolidWorks, ANSYS and COMSOL, combining CAD with structural and multiphysics simulation.

Rope Braiding Machine:

- Designed and modelled a rope braiding machine, focusing on mechanism design and modelling.

Course projects:

- Numerical analysis, clipper circuit simulation, cantilever beam design and Lucas-Washburn equation simulation.

ACHIEVEMENTS, LEADERSHIP & Volunteering

- **GIKI Sports Society (2022–2023)** - Head of Finance - responsible for budgets and stakeholder logistics.
- **Team Urban GIKI (2022–2023)** - Head of Mechanical Design - led the design work of a national student competition team. Conducted Workshops on SolidWorks
- **4th International Workshop on Reverse Engineering (WRE 2022)** – Host – moderated technical sessions for an international academic audience.
- **Shell Eco-Marathon Pakistan 2021 -** 2nd place – Team Urban - Energy-efficient vehicle design and driving competition.
- **Indus Hospital Karachi –** Volunteer in 2018 - Supported patients through consultations and hospital navigation, provided basic first aid and emergency assistance, and improved patient experience through smooth operational coordination.

LANGUAGES

English: C1 – advanced (professional working language)

German: A2 – currently learning, goal is professional working proficiency

Urdu: Native